nRF Connect SDK:

How Nordic uses Zephyr

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nRF Connect Platform Overview

nRF Connect

nRF Connect SDK
- Samples and Applications
- Zephyr RTOS
- Middleware
- Libraries
- Hardware Drivers

nRF Connect for Desktop
- Quickstart
- Bluetooth Low Energy
- Programmer
- Cellular Monitor
- Power Profiler
- nPM PowerUP

nRF Connect for VS Code
- nRF Terminal
- nRF Debug
- Memory Report
- Command Line and GUI
- Create New Board Wizard

Mobile Applications
- nRF Connect for Mobile
- nRF Mesh
- nRF Toolbox
- nRF Blinky
- Device Manager

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nRF Connect SDK Code Base

Contains app code, connectivity protocols, wireless stacks and peripheral drivers

Code is organized in different repositories (Nordic and Open Source (OS) code):

- nrf – Application & Connectivity Protocols
- nrfxlib – Compiled libraries
- nrfx – Peripheral Drivers
- zephyr – RTOS & Board Configuration (OS)
- mcuboot – Secure Bootloader (OS)
- Other Repositories:
  - Trusted Firmware-M, Matter, etc.

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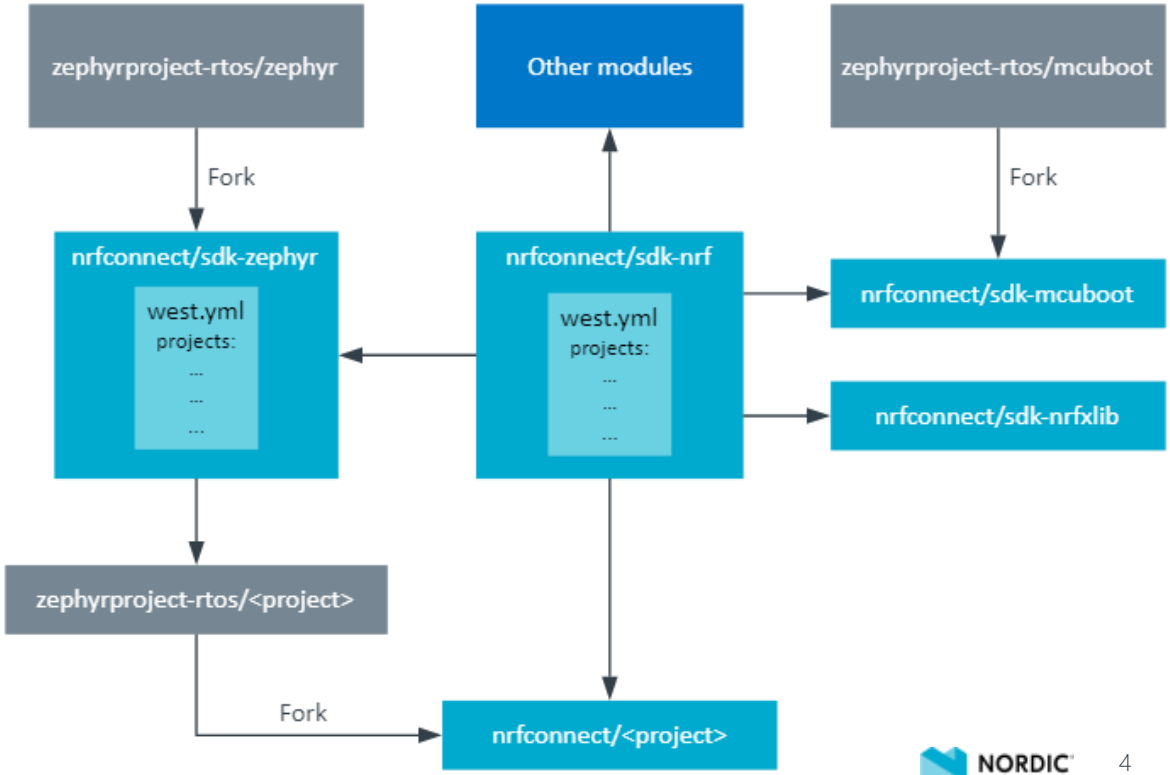
nRF Connect SDK (NCS) Repository Structure

sdk-nrf repository contains the file west.yml in its root folder, which lists all other repositories included in the NCS.

The NCS repository structure has a star topology, with the sdk-nrf repository being the center of the star.



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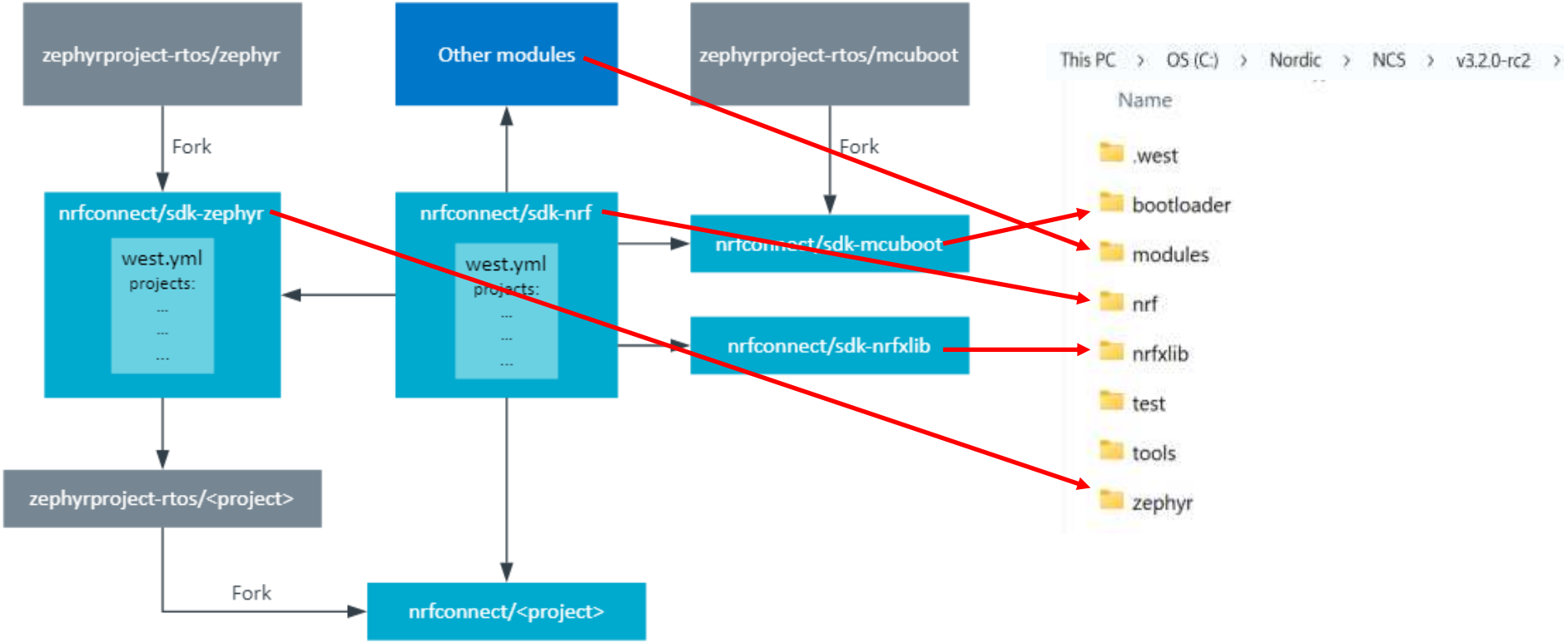
Repository Structure vs. Installation Paths



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Code Examples

Code examples can be found in various repositories, such as ***nrf***, ***zephyr***, or others.

NCS › v3.2.0-rc2 › nrf › samples ›

Name

- app_event_manager
- app_event_manager_profiler_tracer
- app_jwt
- basic
- benchmarks
- bluetooth
- bootloader
- caf
- caf_sensor_manager
- cellular
- common
- crypto
- debug
- dect

NCS › v3.2.0-rc2 › zephyr › samples ›

Name

- application_development
- arch
- basic
- bluetooth
- boards
- cpp
- data_structures
- drivers
- hello_world
- kernel
- modules
- net
- philosophers
- posix
- psa

Nordic TechDocs → docs.nordicsemi.com

Website where documentation can be found, such as device data sheets, certification IDs, dev kit manual, etc.

With regard to SDK, the following can be found, for example:

- nRF Connect SDK => docu of different repositories (sdk_nrf, sdk_zephyr, sdk_mcuboot, …)
- APIs => API docu
- Add-ons => specific west workspaces
- Kconfig Search => Kconfig symbol descriptions
- Software Tools => PC Developement tools, e.g. nRF Connect for VS Code

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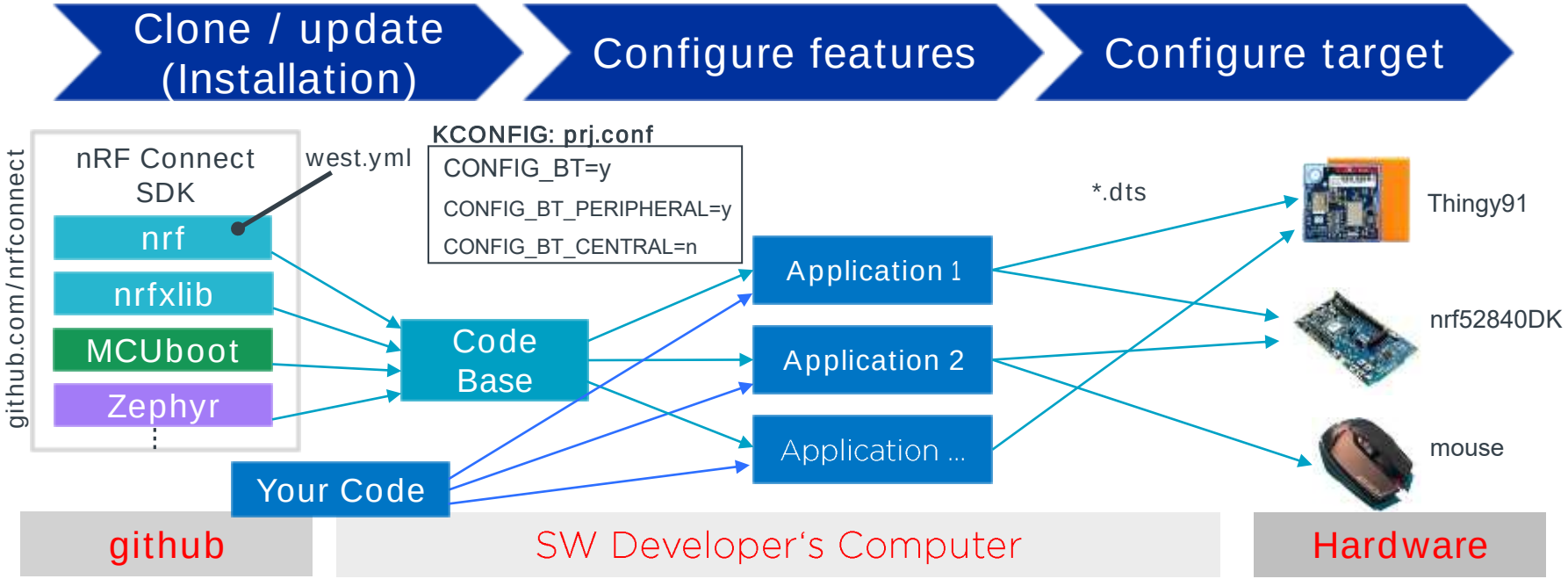
Manage Source Code and Configurations

West
Multi-repository
management tool

Kconfig
Source module / feature
configuration for compile

Device Tree
Target Board / Device
description



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Zephyr Application Types

Repository

An app located in Zephyr repository is referred to as a *Zephyr Repository Application* (e.g. hello_world sample)

```
zephyrproject/
├── .west/
│   └── config
└── zephyr/
    ├── arch/
    ├── boards/
    ├── cmake/
    ├── samples/
    │   ├── hello_world/
    │   └── ...
    ├── tests/
    └── ...
```

West Workspace

An app located within a workspace but outside the Zephyr repository is referred to as a *Zephyr Workspace Application*

```
zephyrproject/
├── .west/
│   └── config
├── zephyr/
├── bootloader/
├── modules/
├── tools/
├── <vendor/private-repos>/
└── applications/
    └── app/
```

Freestanding

Zephyr Freestanding Application is an app located outside a Zephyr workspace (e.g. nRF Connect SDK installation)

```
<home>/
├── zephyrproject/
│   ├── .west/
│   │   └── config
│   ├── zephyr/
│   ├── bootloader/
│   ├── modules/
│   └── ...
│
└── app/
    ├── CMakeLists.txt
    ├── prj.conf
    └── src/
        └── main.c
```

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Zephyr/NCS Application (minimal project)

```
Nordic
├── Workspace
│   ├── MyProject
│   │   ├── CMakeLists.txt
│   │   ├── prj.conf
│   │   └── src
│   │       └── main.c
│   │
```

CMAKE

File: CMakeLists.txt

```
cmake_minimum_required(VERSION 3.20.0)
find_package(Zephyr REQUIRED HINTS $ENV{ZEPHYR_BASE})
project(hello_world)
target_sources(app PRIVATE src/main.c)
```

KCONFIG

File: prj.conf

```
# nothing here! (default settings, e.g. CONFIG_GPIO=y)
```

Source Files

Application (user code) + selected software modules from Code Base (SDK)

File: src/main.c

```
#include <zephyr/kernel.h>

int main(void){
    printk("Hello World!\n");
    return 0;
}
```

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Zephyr/NCS Application – optional parts

```
Nordic
├── Workspace
│   ├── MyProject
│   │   ├── CMakeLists.txt
│   │   ├── prj.conf
│   │   ├── Kconfig
│   │   ├── sysbuild.conf
│   │   ├── VERSION
│   │   ├── board
│   │   ├── snippets
│   │   ├── sysbuild
│   │   └── src
│   │       └── main.c
│   │
├── MyProject_RISCV
│   ├── CMakeLists.txt
│   …
│
├── ncs
│   ├── bootloader
│   │   ├── mcuboot
```

- **Kconfig**
  Configuration System. Allows user defined Kconfig symbols for custom application (used in *.conf files).
- **VERSION**
  Application Version Management. Version is accessible from Source Code, Kconfig, Cmake, and Shell.
- **Board**
  Overlay DTS file allows to change pre-defined settings of a board. Or an own board can be defined from scratch.
- **Snippets**
  save common configuration separately when it applies to multiple different applications.
- **Sysbuild.conf + sysbuild folder**
  used to combine multiple project builds together, e.g. build application together with MCUboot.

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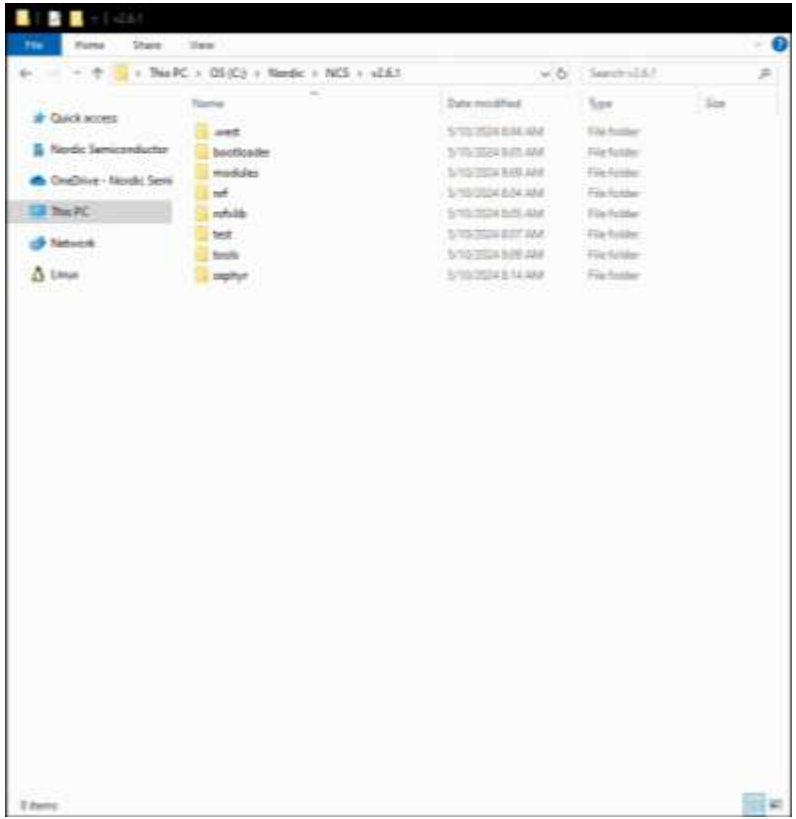
More than 1250 Boards and Shields

- Semiconductors
  e.g. Nordic Semi

- Module Makers
  e.g. Panasonic

- SOM Makers
  e.g. actinius

- Product Makers
  e.g. BBC Microbit



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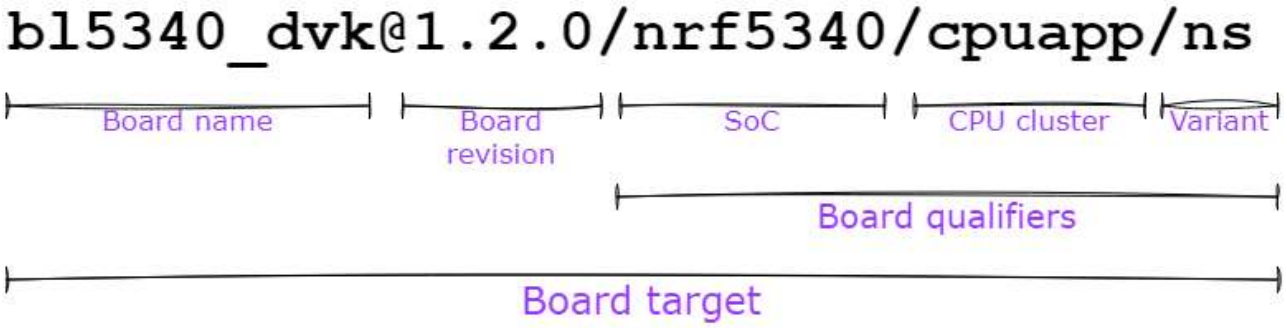
Board Terminology Diagram



board.yml - Notepad

File Edit Format View Help

```
board:
  name: bl5340_dvk
  vendor: lairdconnect
  socs:
  - name: 'nrf5340'
    variants:
    - name: 'ns'
      cpucluster: 'cpuapp'
```

Ln 100% Windows (CRLF) UTF-8

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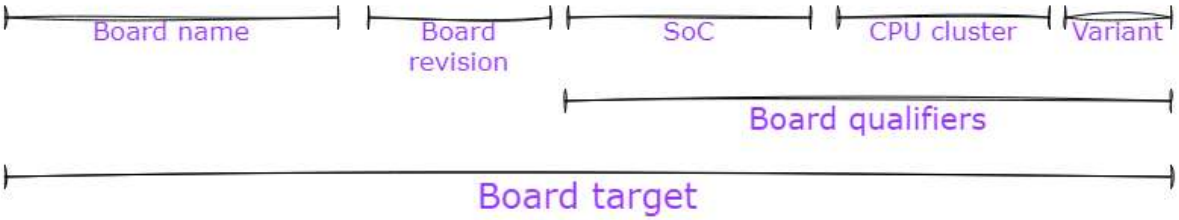
Board Terminology Diagram

nrf9160dk@0.14.0 / nrf9160 / ns



nrf9160dk / nrf9160 / ns

board.yml - Notepad

File Edit Format View Help

```
board:
  name: nrf9160dk
  vendor: nordic
  socs:
  - name: nrf9160
    variants:
    - name: 'ns'
  - name: nrf52840
  revision:
    format: major.minor.patch
    default: "0.14.0"
    revisions:
    - name: "0.7.0"
    - name: "0.14.0"
```

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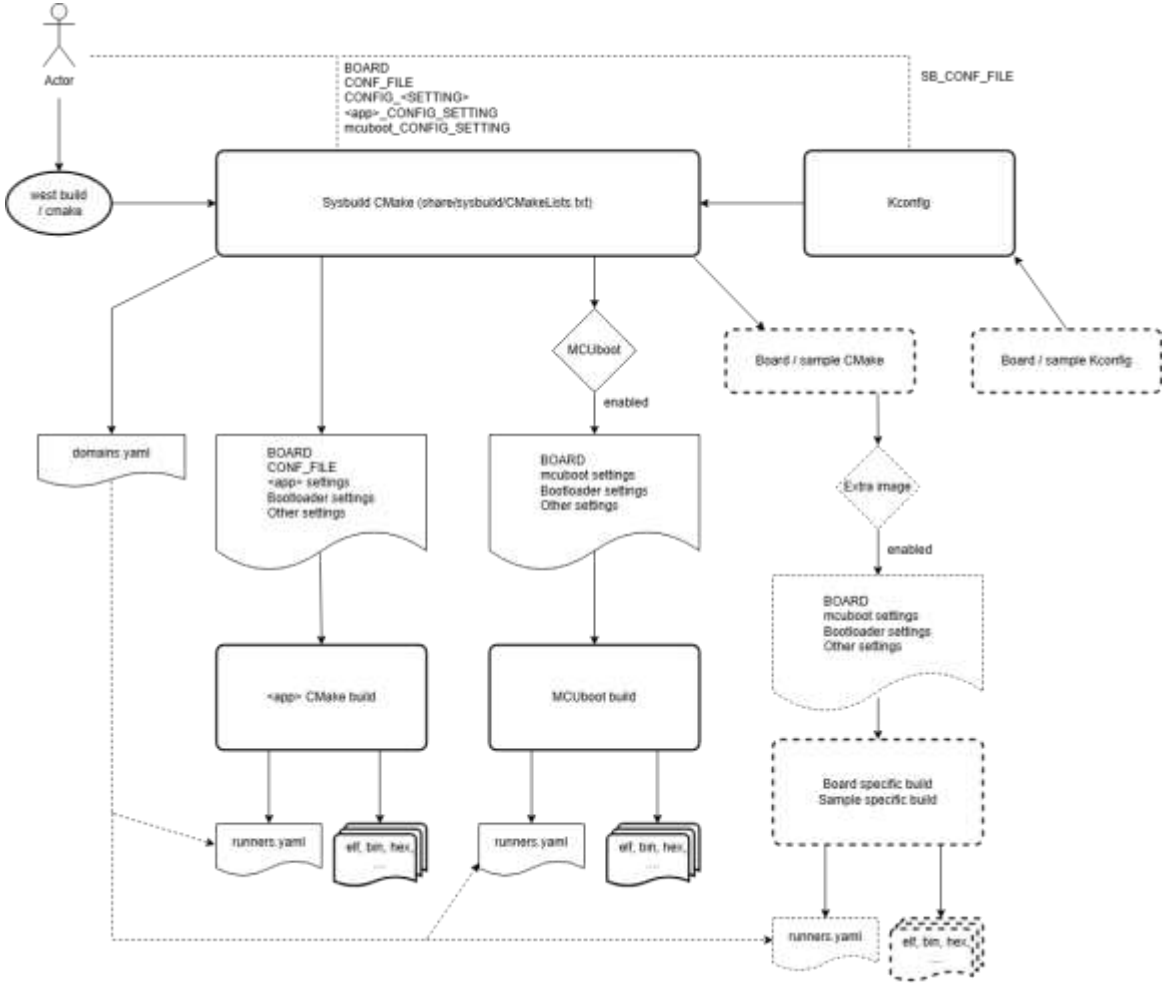
SYSBUILD



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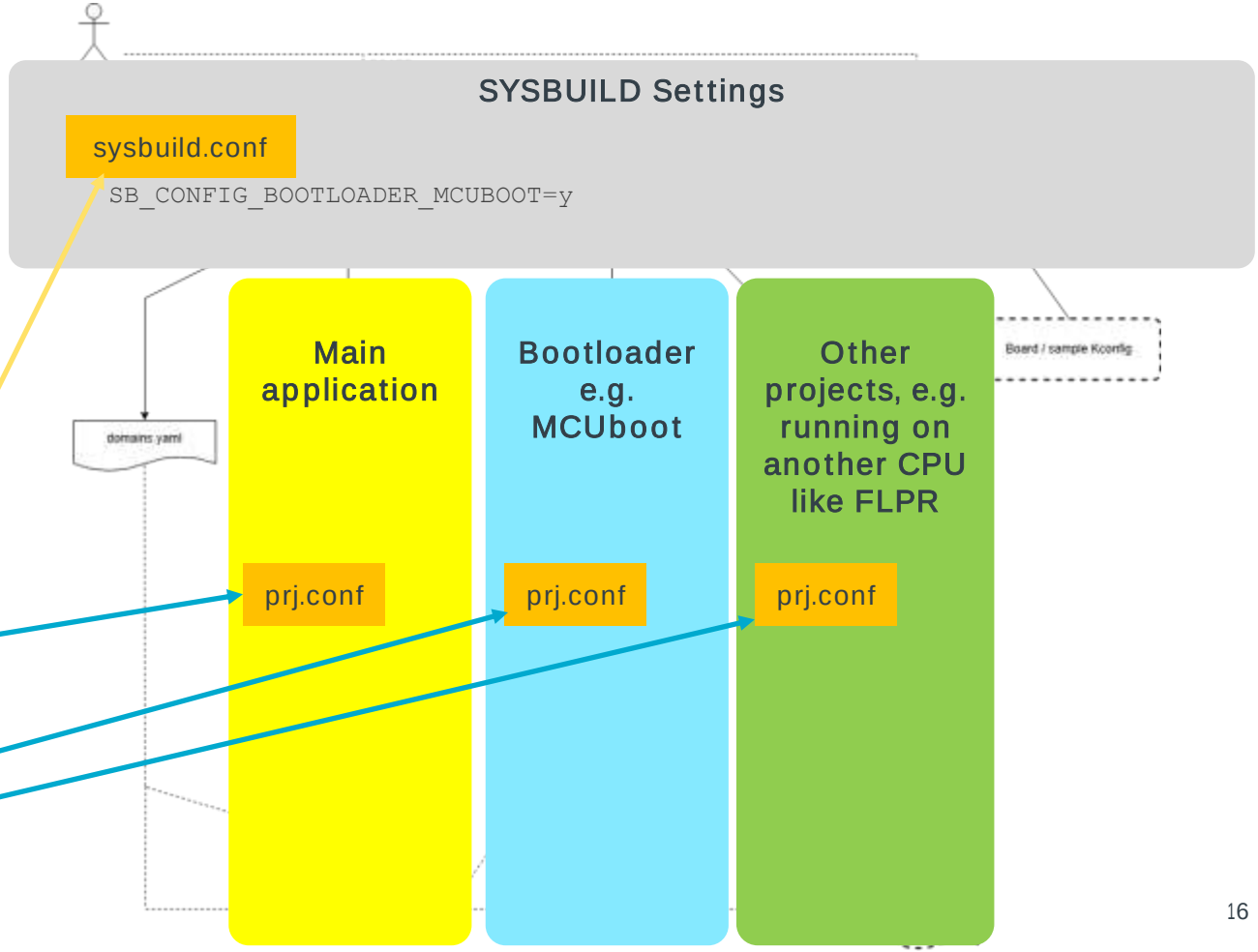
SYSBUILD




```
<home>/application
├── src
│   └── main.c
├── CMakeLists.txt
├── prj.conf
├── sysbuild.conf
└── sysbuild
    ├── mcuboot.conf
    └── my_sample.conf
```